

Birla Institute of Technology & Science, Pilani

Work Integrated Learning Programmes Division

QUESTION BANK

Course Title: DEEP NEURAL NETWORK

Course Code: AIMLCZG511

Important Announcement

This question bank is only for practice and revision. It does not guarantee that any of these questions, or similar questions, will appear in the examination. Please do not prepare only from this question bank.

For the exam, prepare the complete course: revise the lecture slides, practise the numerical problems discussed in class, and focus on conceptual and case-study/application-based questions. Be prepared to apply the concepts to new situations and changed numerical values.

Preparation Guidelines

- Revise all lecture slides and examples discussed in class.
- Practise all numerical problems discussed during lectures.
- Focus on understanding concepts, not memorizing questions.
- Practise applying concepts to new case studies and situations.
- Prepare the complete course content; this bank is only a practice resource.

Good Luck for the Exam!

Wishing you all the very best for your examination.

Q1. A bank uses a single-layer neural network to classify loan applications based on income and credit score. During testing, the classifier consistently misclassifies cases where the positive and negative examples cannot be separated by a straight line. Analyze the reason for this behavior and explain what architectural change would make the problem solvable.

Q2. A development team increases the number of epochs of a single-layer Perceptron because the model is unable to correctly classify a particular dataset. Explain whether increasing training iterations alone can overcome the limitation of the model.

Q3. Consider a neuron with $x = [1, -2]$, $w = [0.6, -0.4]$, and $b = 0.2$. Calculate the net input and output using an appropriate activation function.

Q4. A classification system initially uses a Perceptron but later requires learning nonlinear decision boundaries. Compare the role of the input, hidden and output layers in the redesigned network.

Q5. A company replaces a shallow network with a deeper network for image recognition. The training accuracy improves significantly, but the engineers are unable to explain why. Explain how depth enables hierarchical feature learning using a suitable example.

Q6. A medical-screening model performs well when the classes are linearly separable but fails when several interacting features determine the class. Explain how adding hidden neurons can transform the representation learned by the model.

Q7. A team argues that a sufficiently large number of neurons in a single-layer Perceptron should be able to solve any binary classification problem. Evaluate this claim using a suitable decision-boundary example.

Q8. A neural network uses Sigmoid activations in every hidden layer. As the network becomes deeper, the gradients reaching the first hidden layer become extremely small. Explain the phenomenon and recommend a more suitable activation strategy.

Q9. For a neuron with $z = -2$, calculate the output of Sigmoid, ReLU, and Leaky ReLU with $\alpha = 0.01$. Compare their behavior.

Q10. For $z = 2$, calculate the Sigmoid activation and its derivative. Explain what the derivative indicates about gradient propagation.

Q11. A deep network uses Tanh activations in several hidden layers. The training process becomes slow even though the loss decreases. Analyze whether Tanh could contribute to the problem.

Q12. A computer-vision model replaces Sigmoid hidden units with ReLU. The model trains faster, but some neurons stop responding completely. Identify the possible problem and suggest a modification.

Q13. A regression model requires outputs that can take both positive and negative values. Compare the suitability of ReLU, Sigmoid and Tanh for the output layer and justify a choice.

Q14. Two hidden-layer designs use ReLU and Leaky ReLU respectively. A developer reports that some units in the ReLU network permanently output zero. Explain why this can happen and how the alternative activation changes the behavior.

Q15. A predictive-maintenance network has two input neurons, two hidden Sigmoid neurons and one Sigmoid output neuron. Given the input, weights and biases, perform complete forward propagation

Deep Neural Network | AIMLCZG511 | Practice Question Bank | Page 3 and calculate the final prediction.

- Q16.** For a binary classification network using Binary Cross-Entropy loss, derive the output-layer error term using the chain rule and show why it simplifies to $\delta^{(2)} = \hat{y} - y$.
- Q17.** A neural network predicts a probability of 0.25 for a sample whose actual class is 1. Calculate the Binary Cross-Entropy loss and interpret the result.
- Q18.** During backpropagation, the output neuron produces a negative error term. Explain what this indicates about the current prediction and the direction in which the output-layer weights are likely to move.
- Q19.** A student calculates the output-layer gradient correctly but obtains an incorrect hidden-layer gradient. Identify the role of the chain rule in propagating the error backward and explain where activation derivatives enter the calculation.
- Q20.** A two-layer classifier gives a confident but incorrect prediction. Explain how the loss and output error signal would respond and why this can produce a comparatively large corrective update. Deep Neural Network | AIMLCZG511 | Practice Question Bank | Page 3
- Q21.** An engineer changes the activation function of a hidden layer but keeps the same weights and inputs. Explain which quantities in the forward pass and backpropagation would change and why.
- Q22.** A neural-network parameter has $w = 0.8$, learning rate 0.1, and gradient -0.6 . Calculate the updated parameter and explain why the parameter increases despite gradient descent.
- Q23.** A training dataset produces gradients 0.4, 0.6, 0.2, and 0.8. Calculate the parameter update using Batch Gradient Descent, SGD using the first sample, and Mini-Batch GD using the first two samples.
- Q24.** A company trains a DNN using SGD and notices that the loss fluctuates heavily between iterations. Another implementation using Batch GD is extremely slow. Recommend a training strategy and justify your decision.
- Q25.** A model's loss decreases very slowly after several epochs. The developer proposes increasing the learning rate by 20 times. Analyze the possible benefits and risks of this decision.
- Q26.** Two networks use learning rates of 0.001 and 1.0 respectively. Without performing numerical calculations, explain how their training behavior could differ.
- Q27.** A training run reaches a low loss initially but then becomes unstable immediately after the learning rate is increased. Diagnose the likely reason and suggest a more appropriate learning-rate strategy.
- Q28.** Two teams use identical networks and datasets but obtain different convergence behavior because one uses full-batch updates and the other uses mini-batches. Explain how the update frequency and gradient estimates can account for the difference.
- Q29.** A 10-layer network receives an output gradient of 0.8. The gradient multiplier at every layer is 0.25. Calculate the approximate gradient reaching the first layer.

Q30. Repeat the previous analysis when the gradient multiplier is 1.8. Identify the training problem.

Deep Neural Network | AIMLCZG511 | Practice Question Bank | Page 4

Q31. An engineer observes that the final layers of a DNN learn rapidly while the first few layers hardly change. Diagnose the likely gradient problem and explain why it occurs.

Q32. A deep network suddenly produces extremely large parameter updates and eventually generates NaN values during training. Identify the likely cause and suggest two remedies.

Q33. A research group is choosing between Xavier initialization, He initialization, Batch Normalization, Residual Connections and Gradient Clipping. For a network suffering from unstable gradients, select two techniques and explain why they are appropriate.

Q34. A network using many Sigmoid layers shows almost no change in its early-layer weights. Explain how repeated multiplication of derivatives can create this behavior.

Q35. A deep network's gradients become very large only for a small number of training batches. Explain why gradient clipping could be useful and distinguish its purpose from that of careful weight initialization.

Q36. A CNN receives a $28 \times 28 \times 1$ image. It contains a 3×3 convolution with 8 filters, stride 1 and no padding, followed by 2×2 max pooling with stride 2. Calculate the output dimensions after both layers.

Q37. A CNN receives an RGB image of size $64 \times 64 \times 3$. A convolution layer contains 16 filters of size 3×3 , stride 1 and padding 1. Calculate its output size and number of trainable parameters.

Q38. A CNN has a convolution layer with 32 filters of size 5×5 , receiving 16 input channels. Calculate the number of trainable parameters including biases.

Q39. A student claims that pooling layers increase the number of trainable parameters because they introduce additional operations. Evaluate the claim.

Q40. A surveillance company considers two models: one that directly flattens a $128 \times 128 \times 3$ image and another that first applies convolution and pooling. Explain why the second approach is generally more suitable for image analysis.

Q41. A CNN receives a $32 \times 32 \times 3$ image and applies a 3×3 convolution with stride 2 and padding 1. Determine the output dimensions and explain how changing the stride affects spatial resolution.

Q42. A model designer doubles the number of filters in a convolutional layer while keeping the kernel size unchanged. Explain which dimensions of the parameter count and feature representation are affected. Deep Neural Network | AIMLCZG511 | Practice Question Bank | Page 4

Q43. A CNN contains a sequence of 3×3 convolutions and 2×2 pooling layers. Determine the receptive field of a neuron in a later feature map with respect to the original image. Clearly show the calculation layer by layer.

Q44. A CNN designer replaces one 7×7 convolution with three consecutive 3×3 convolutions. Analyze the effect on receptive field, parameters and number of nonlinear transformations.

Q45. A company chooses between LeNet-5 and VGG-style architecture for a complex image-classification problem. Explain why the deeper architecture may learn richer representations.

Q46. A VGG-style CNN receives a $224 \times 224 \times 3$ image and applies three 2×2 max-pooling operations with stride 2. Determine the spatial dimensions before the fully connected layer. Deep Neural Network | AIMLCZG511 | Practice Question Bank | Page 5

Q47. An autonomous-vehicle vision system requires detection of increasingly complex features—from edges to textures to object parts. Explain how a CNN's hierarchical feature extraction enables this progression. Compare this with directly connecting the pixels to a fully connected layer.

Q48. A CNN designer observes that increasing the stride of an early convolution reduces the size of later feature maps. Explain the trade-off between computational efficiency, spatial detail and receptive field.

Q49. An image-classification model performs well on large objects but poorly on small defects. Analyze how pooling, stride and receptive-field growth could contribute to this behavior and suggest an architectural adjustment. Deep Neural Network | AIMLCZG511 | Practice Question Bank | Page 5 Deep Neural Network | AIMLCZG511 | Practice Question Bank | Page 6